Software Modelling and Design

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Requirements analysis

Requirement definition

- **Requirement**: Description of a program / what it must do.
- A requirement must be 'verifiable'. In other words, there must be some way to show that the system follows a requirement.

Types of requirements

Functional requirement

- What does the program actually do for the user?
- It should focus on 'what', not 'how' and the intended purpose of the system
- Example:
    - "It must allow the user to log in using a login page"

Nonfunctional requirement

- Qualities of the program; it doesn't require it to function properly.
- Example:
    - "It must be multi-platform"

Design requirement

- Defines how the system will operate or look.
- Example:
    - "It will use a MySQL database to store users' details"
- DO NOT IDENTIFY THESE.
- They will constrain the design of the system.

Constraints

- A global requirement that applies everywhere to limit the freedom you have with your system.
- Example:
    - "Must be able to function on at least 1GB of RAM"

Assumptions

- Properties of the environment that are needed for the system to function properly. <u>The software cannot control it.</u>
- Another name for assumptions are 'facts' or 'statements of facts'.
- Example:
    - "Users keep their passwords secret"
    - "If brakes are applied, it takes 3 seconds to stop"

Types of functional requirements

Normal

- Regular functionality of the system
- Example:
    - “Users can send emails”

Error

- Functionality that handles things when something goes wrong
- Example:
    - “If there is a failure, make a log file”

Safety

- Ensures safety of the user
- Example:
    - “Do not move the elevator when the doors are open”

Security

- Functionality that enforces security with things like user data.
- Example:
    - “Hash passwords; do not store them raw”

Requirement definition process

Define scope

- This is the overall picture; what are you trying to achieve?
- You need to define your:
    - Need - Why are you making this system?
    - Goals - What functionality will it provide to stakeholders?
    - Business Case - What will the business gain from this system?
    - High-level operational concepts - How will the product be used?
    - Stakeholders - Who will be involved in the development/usage/maintenance of the system?
- Example:
    - Need - We’re losing business to competitors
    - Goals - To provide services for customers
    - Business Case - The business will stay in competition
    - High-level operational concepts - Customers search and buy items
    - Stakeholders - Customers, marketing department, IT department etc.

Operational scenarios

- Define a bunch of case studies for the system.
- They should describe the use of the system.
- Each case study should be a series of interactions.
- From these case studies, you can quickly gain an understanding of what's happening.
- It will also help generate requirements and avoid leaving ones out.
- You can derive case studies by:
    - using similar systems
    - asking the marketing department
    - asking customers
- Example:
    - User logs in with user ID and password
    - Gets overview of their account
    - Selects current account and is shown transactions

- There is a difference between a scenario and a use case:
    - **Use case**: A list of actions or event steps defining the interactions between a user and the system to achieve a goal. It covers pretty much every possibility.
    - **Scenario**: A documentation of a foreseeable series of interactions between the user roles and the system to achieve a goal. Think of it like an instantiation of a use case, like how an object is an instantiation of a class.

- A use case has multiple components:
    - **Use case name** - The name of the use case
    - **Participating actors** - The stakeholders that participate in this use case
    - **Entry condition** - The first thing that happens that fires off the use case
    - **Flow of events** - An ordered set of events executed by the system and the actors.
    - **Exit conditions** - What happens at the end of the use case.
    - **Exceptions** - Describe all the exceptions of the use case.
    - **Quality Requirements** - Requirements that are not related to the functionality of the system.

Identify interfaces

- How will your stakeholders interact with the system?
- Example:
    - Customers:
        - PC, phone, tablet, browser
        - Smartphone app
    - Bank staff:
        - Branch IT access system
    - Admin:
        - Database management system (e.g. phpMyAdmin)

        - Server

Writing requirements

- This is where you split up your documentation into individual requirements.
- Each requirement must be:
    - simple, concise and be a simple thought
    - unambiguous (be clear)
- Distinctions:
    - Functional: “The system shall”
    - Assumptions: “The user will”
- Labels (to identify each requirement):
    - ASM1, ASM2, ASM3 ... are assumptions
    - FUN1, FUN2, FUN3 ... are functional requirements

Level requirements

- Functional requirements can be further split up into different labels, with things like authentication, accounts, services etc. depending on your project.
- Example:
    - UACC - Users and accounts
    - AUTH - Authentication
    - SERV - User services
- Applied example:
    - UACC1 (FUN1a): The system shall maintain user IDs
    - AUTH1 (FUN1b): The system shall maintain a password with an ID
    - AUTH2 (FUN2): The system shall provide a login page with password and ID input boxes.
    - UACC4 (FUN11): The system should ensure that the bank balance does not go into negative
    - SERV1 (FUN9): At the start of a login session, the system displays a list of account numbers

Access verification

- How will you show that a requirement has been met?
- You can show it in three different ways:
    - Design review (literally just look at the system to see if it fits the requirement)
    - Test scenario (like a case study earlier on)
    - Additional functionality (kinda like JUnit)
- Example:
    - The system shall maintain user IDs:
        - Add a function for the tester to view a list of user IDs
    - The system shall maintain passwords with IDs
        - Inspect the design of the database
    - The system should provide a login page with password and ID input boxes
        - Use a test scenario to log in

Access verification (sanity check)

- Are the requirements:
    - necessary?
    - verifiable?
    - attainable?
    - clear and understandable?
    - free from implementation bias?
- If not, rethink that requirement.

Baseline requirements

- Just talk to everyone to make sure everyone understands and agrees with the requirements.

UML

Code design

Why use UML

- UML is a form of **abstraction**.
- Abstraction: only showing the important information; leaving out unimportant stuff.
- Example:
    - UML only shows class hierarchies; does not show implementation

Models

- A model is a form of abstraction that represents something.
- Models can help us visualise and validate requirements before actually implementing it.
- Models are also a good way of showing others how the system will work.
- Models can also help spot problems early.
- There are 4 types of model:
- **Functional model**: What are the functions of the system (what does it do)?
- **Object model**: What is the structure of the system (how is it laid out)?
- **Dynamic model**: How does the system react to external events (how does it react to things)?
- **System model**: Object + functional + dynamic (All of the above models put into one)

Design representation

- There are three kinds of design representation:
- **Textual**: Representing things in pure text
- **Diagrammatical**: Using a diagram to represent something
- **Mathematical**: Using maths to represent something

Dealing with complex problems

- There are three ways to deal with complex problems:
- **Divide and conquer**: Split problem up into sub-problems that can all be solved using the same bit of code (common in recursion)
- **Functional decomposition**: Split problem up into modules
- **Object-oriented decomposition**: Split problem up into classes

Hierarchy relationships

- There are two kinds of relationships in hierarchy:
- **Part-of hierarchy**
    - Like an attribute; an instance of one ‘has’ an instance of another
    - Example:

        - A CPU has registers
        - A ‘Graph’ has a ‘List’ of nodes
- **Is-kind-of hierarchy** (taxonomy)
    - Like inheritance; one is a parent, and the other is a child
    - Example:
        - A blood cell is a type of cell
        - A dog is a type of animal

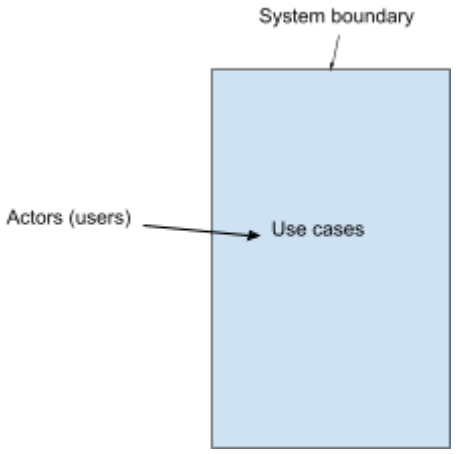
Modelling in UML

Functional modelling (use case diagrams)

- Functional modelling models what the system does for the user.
- A use case diagram is a generalised description of how a system will be used
- Provides an overview of the functionality of the system
- Represents stakeholders interacting with the system in different ways
- Shows **what** the system will do, not **how** it will do it

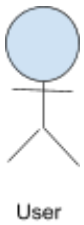
System boundary

- A system boundary, or a ‘subject’, is a rectangle separating stakeholders and the system. It encapsulates the system away from the actors/users.



Actors

- A stakeholder interacting with the system. Can be a stickman shape
- Example:
    - Customer
    - Admin

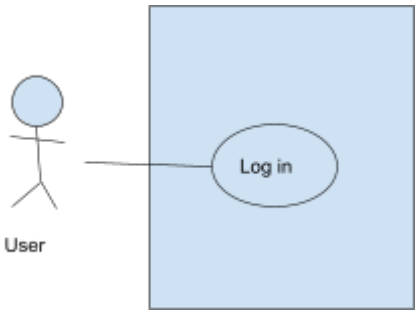


Use case

- A bubble that represents an interaction
- Example:
    - Log in
    - Register
    - Place an order

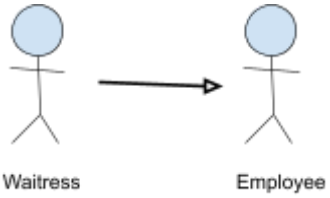
Association

- Communication between actor and use case; represented by a solid line.
- Example:
    - Customer --- (Log in) **The customer can log in**
    - Admin --- (Change price of item) **The admin can change the price of an item**



Generalisation

- Relationship between one general element to a more specific element.
- Represented by an arrow with a triangle at the end of it
- Example:
    - Waitress → Employee **Waitress 'is an' Employee**

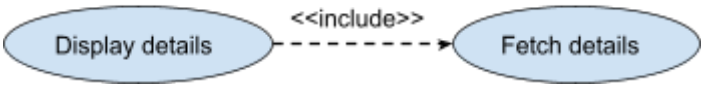


Types of generalisation

Include

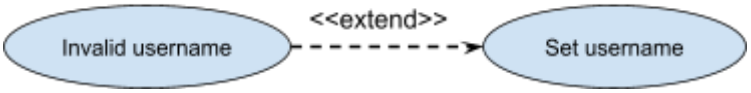
- When a use case uses another use case to function, it has to <<include>> it.
- We do this so we don't have repeated code in two use cases.
- Represented with <<include>> above generalisation arrow.
- Example:

- (modify orders) -- <<include>> → (list order)
- (display details) -- <<include>> → (fetch details)



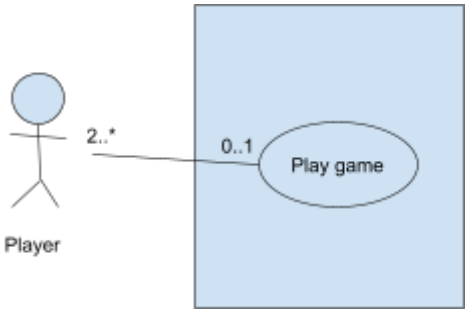
Extend

- A use case adds optional behaviour to another use case by extending it.
- We do this to add optional behaviour which only occurs under certain circumstances (for example on errors/exceptions)
- Represented with <<extend>> above generalisation arrow.
- Example:
  - (Invalid username) -- <<extend>> → (Set username)

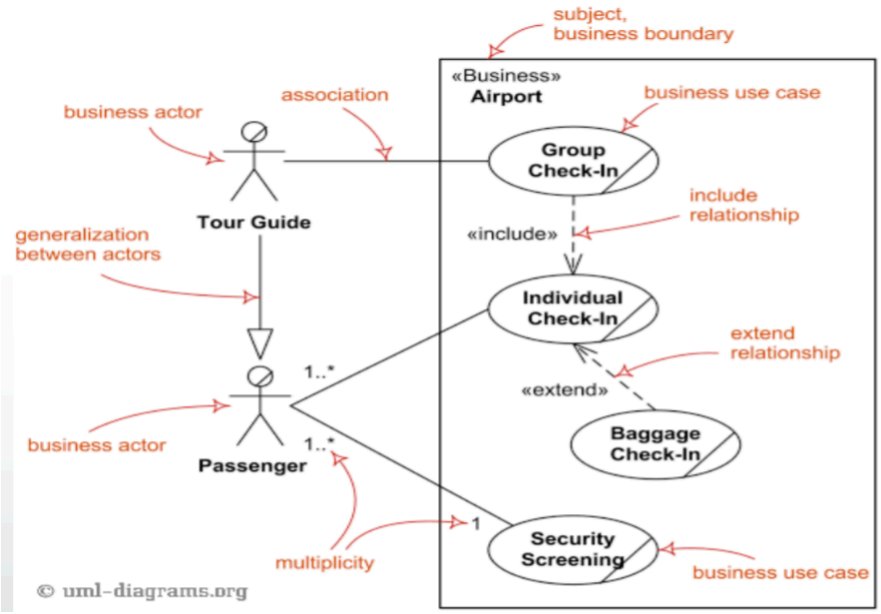


Multiplicity

- A number range indicating one-to-one, one-to-many, many-to-one or many-to-many relationships.
- Notation:
  - 1 One and only one
  - 0..1 Zero or one
  - M..N From M to N
  - * From zero to any positive integer (use this instead of 0..*)
  - 0..* From zero to any positive integer (use * instead)
  - 1..* From one to any positive integer
- Example:
  - Player 2..* ---------- 0..1 (Play game)
    - 0..1 = A player can either be playing a game or not playing a game (possibly in the lobby room)
    - 2..* = A game must have 2 or more players.



Use case diagram example

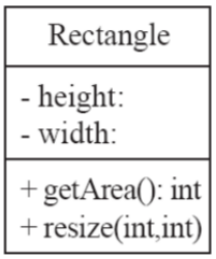


Object modelling (class diagrams)

- Object modelling models the actual classes and entities used in the system.
- Class diagrams partition the system into areas of responsibility (classes) and shows associations between them.
- Can be presented at three distinct levels:
    - **Conceptual** - Represents concepts in project domain
    - **Specification** - Shows interfaces between components
    - **Implementation** - Shows classes that correspond directly to code

Classes

- A class is represented as a box split in three with a **name**, **attributes** and **methods**.
- The name goes at the top, the attributes go below the name and the methods go below the attributes.
- '+' means public, '#' means protected, '~' means default and '-' means private.
- Whenever you see ':' followed by a word, that means type. For methods, that's a return type. For attributes, that's a variable type (e.g. getArea() : int means that getArea() returns 'int').
- Example:



Associations and multiplicity

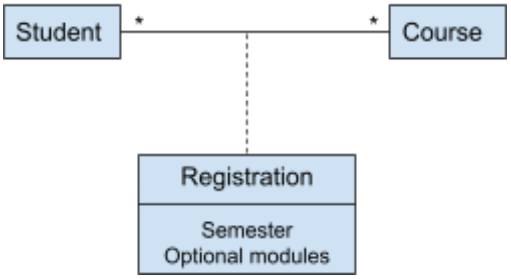
- Multiplicity works the same way as use case diagrams

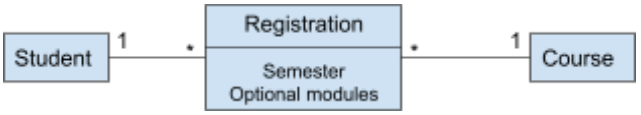
Labels

- You can label an association to be more specific about what the association entails.

Association classes

- Let's just say you have students and they're studying courses. That's simple; a course can have many students, and a student can study many courses.
- But now you want to store what <u>semester</u> they're in, what <u>optional modules</u> they've picked etc.
- Do we store these attributes in Student or Course? Neither, because they're attributes that describe the association, not the actual classes.
- Therefore, we can have association classes whose main purpose is to describe an association.
- This class will be the 'bridge' between Student and Course. Both of these syntaxes will be valid:



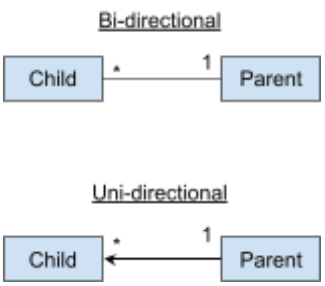


Reflexive associations

- A class can be associated with itself. Think of it like a linked list.
- An example of this would be an employee that manages another employee.

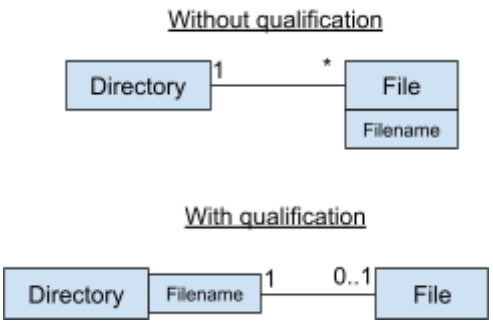
Directionality

- Normally, associations are bi-directional (you can go from one class to another).
- You can also set associations to uni-directional, meaning you can go from one class to another, but not back again.
- In the example, you can reference a parent's child, but you cannot reference a child's parent. This can be thought of as a form of abstraction.



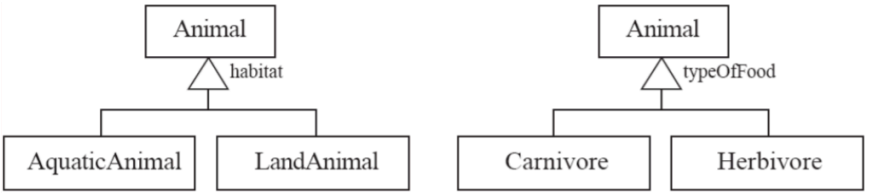
Qualifiers

- Let's say you wanted to model directories with files.
- A directory can have many files, and a file can only belong to one directory.
- Seems simple, right?
- Well, two files in the same directory cannot share a filename.
- How do we model this in UML?
- Given that filename is a property of File, we can make filename a **qualifier**.
- A qualifier is like a primary key; it can't be repeated for a given association.
- This makes the relationship between Directory and File a **qualified association**.



Generalisations

- Generalisations in class diagrams have the same concepts as inheritance in OOP.
- A generalisation set is a labelled group of generalisations with a common superclass.
- The label (sometimes called the discriminator) describes the criteria used in the specialisation (in other words, it describes the inheritance).
- In the example, 'AquaticAnimal', 'LandAnimal', 'Carnivore' and 'Herbivore' are all subclasses, with 'Animal' being the superclass. 'habitat' and 'typeOfFood' are labels.

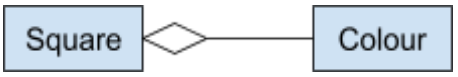


Aggregations and compositions

Aggregation

- Aggregations are special kinds of associations meaning that one class is "a part of" another class.

- They are represented with a hollow diamond at the end of the association line.
- The semantics of an aggregation is the same as a normal association; it mainly exists to help people understand how the system works better.
- Example:
    - You have a Square, and it has a Colour. If you deleted the Square, the 'Colour' may still exist.

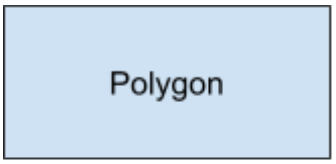


Composition

- A composition is like a strong kind of aggregation, meaning that one class "has" another class.
- They are represented with a filled diamond at the end of the association line.
- Example:
    - You have a ChessBoard with 64 "Place"s, each representing a square on the board. If you deleted the ChessBoard, you would also have to delete all 64 "Place"s as there is no reason for them to exist anymore.
    - You have a Building, and a Building has Rooms. If you destroy the Building, you also destroy the Rooms.

Object Constraint Language (OCL)

- OCL is used to define constraints within class diagrams that are outside the scope of graphical elements.
- It's kind of like a programming language, except you can't create non-boolean values and you cannot modify data.
- Everything made in OCL must be an expression, and it must always be true, hence why it's a 'constraint'.
- OCL always starts with a curly brace and ends with a curly brace.
- OCL's syntax is very similar to languages like C++, Java, C# etc.
- OCL's variable structure isn't important; as long as you express what the constraint is, that's all that matters.
- Example:
    - You have a polygon. You must ensure that the last node of the polygon is in the same place as the first node of the polygon.



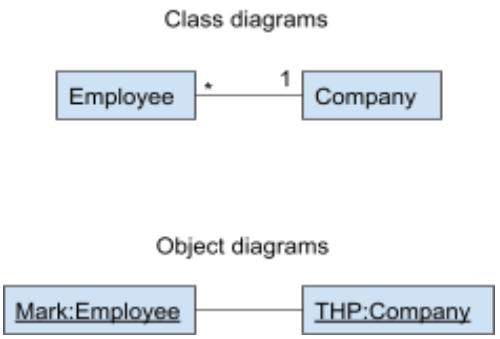

```
{edge.first().position
== edge.last().position}
```

Object modelling (object diagrams)

- An object diagram is like a class diagram, except it has a context. Like how an object is an instance of a class, an object diagram is an instance of a class diagram.
- Think of it like a use case for the class diagram.
- Names of objects are in the format ‘object_name:class_name’, for example, Mark:Employee or THP:Company
- Names of objects are also underlined to show that they are objects.

Things in class diagrams	**Things in object diagrams**
Class	Object
Association	Link
Generalisations	N/A (they don’t exist in object diagrams)

- Example:



Types of objects

- There are three kinds of objects:
- **Entity** objects - represents actual information
- **Boundary** objects - represents interaction between user and system
- **Control** objects - represents control tasks performed by the system
- You can visualise these like in the model-view-controller architecture (MVC):
    - Model <-> Entity
    - View <-> Boundary
    - Controller <-> Control

Abbott’s Heuristic

- Abbott’s Heuristic is a way of converting descriptions into class/object diagrams.
- Since it’s a heuristic, it’s not perfect. Take it with a pinch of salt.

Part of speech	**Model component**	**Examples**
Proper noun	Instance	Alice

Common noun	Class	Field officer
Doing verb	Operation	Creates, submits, selects
Being verb	Inheritance	Is a kind of, is one of either
Having verb	Aggregation	Has, consists of, includes
Modal verbs	Constraints	Must be
Adjective	Attribute	Blue square

Dynamic modelling

- Dynamic modelling models how the system changes and how units communicate with each other.

Interaction diagrams

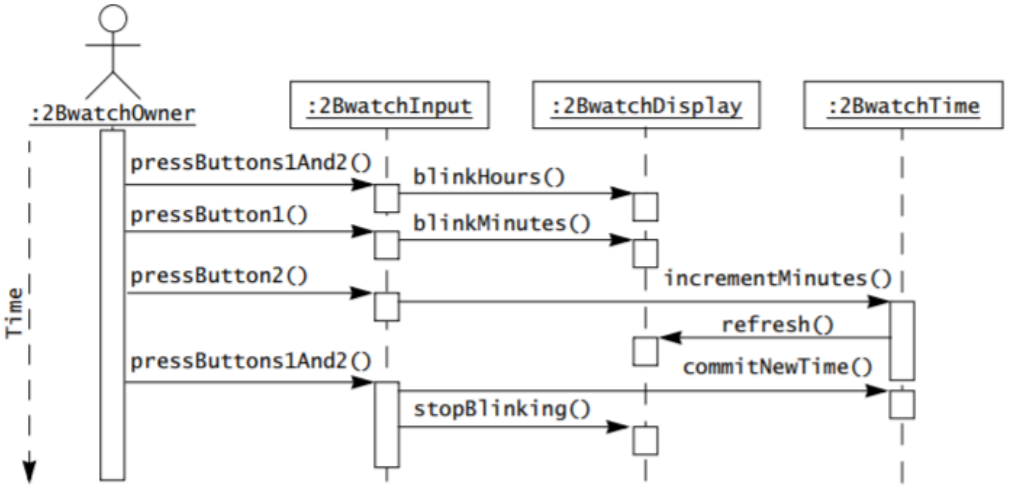
- Interaction diagrams describe patterns of communication between different parts of the system (objects).
- An object can interact with another object by sending a ‘message’.

Sequence diagrams

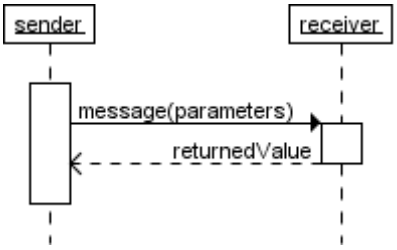
- A sequence diagram shows communication through a sort of “timeline”.
- The horizontal shows interactions, and the vertical shows time (from top to bottom).
- Each class has a vertical timeline, called a ‘lifeline’.

- *There are three kinds of lifelines:*
    - *Phone a friend*
    - *50/50*
    - *Ask the audience*
- *Just kidding; a lifeline can be whatever you want it to be.*

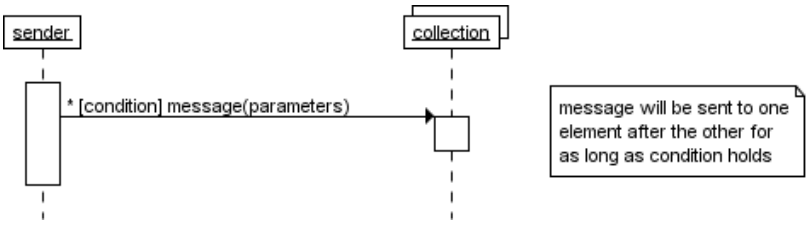
- To follow the diagram, you read from the top and work your way down.
- Every arrow shows an object interacting with another object.
    - The first column should be an **actor**.
    - The second column should be a **boundary object**.
    - The other columns should be **control / entity objects**.
- The user is shown as a stickman (like use case diagrams) on the left, and every boxed object is a part of the system.



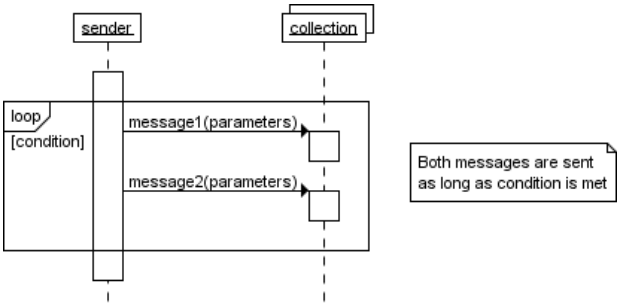
- A dotted line represents ‘dataflow’, which could mean a return value.



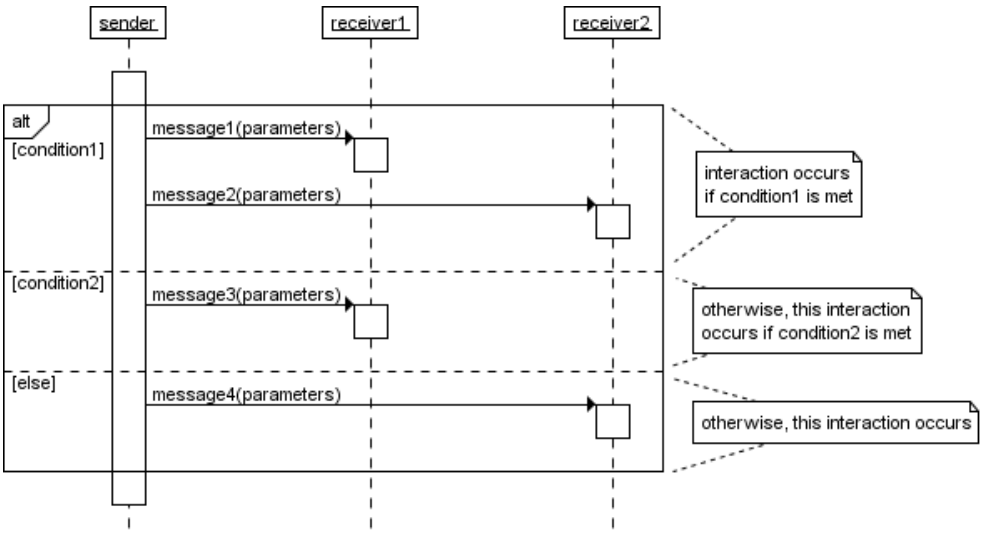
- An asterisk followed by a [condition] works the same way as a while loop; as long as the condition is true, the command will keep on executing in iterations.



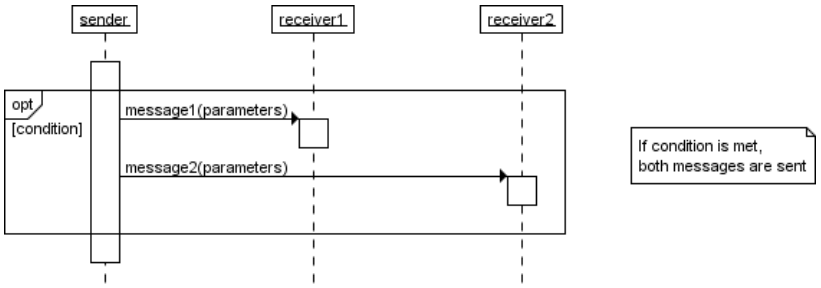
- This can also be achieved with the ‘loop’ combined fragment.



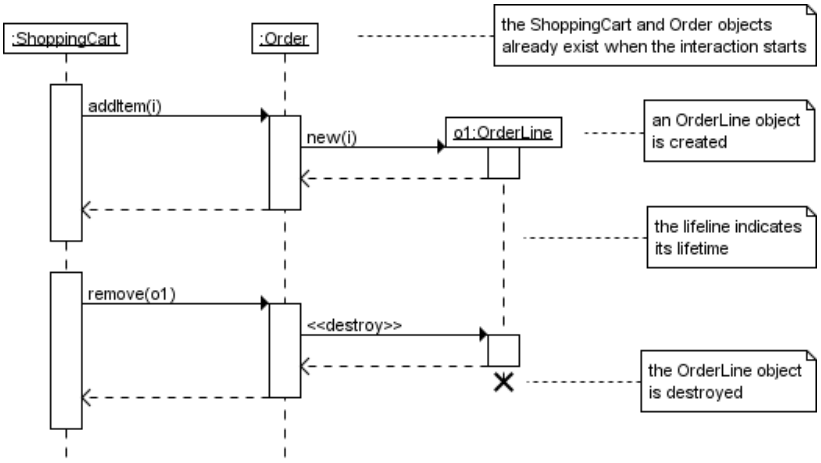
- You can also have if statements using the ‘alt’ and ‘opt’ combined fragments:
    - ‘alt’ is used for when there are alternate scenarios, e.g. pay in credit or cash



- ‘opt’ is used for when there is an optional step, e.g. would you like a bag, yes or no



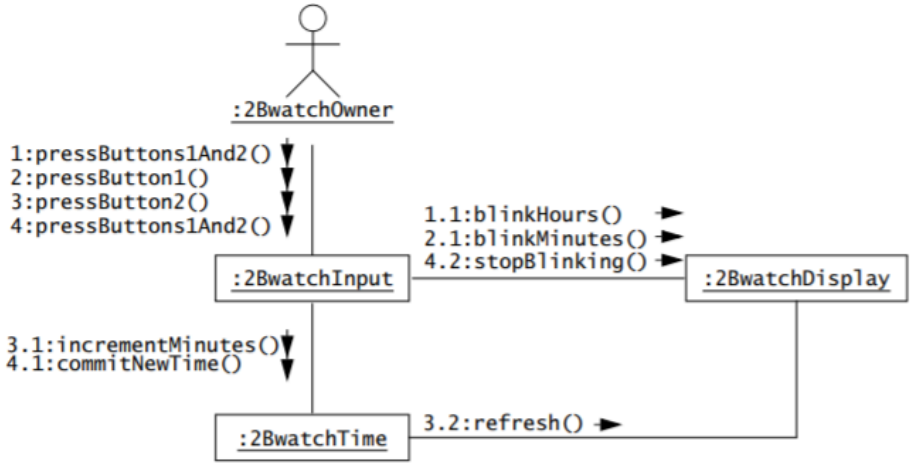
- When an arrow points to a class box, it means ‘creation’. A new instance of that object has been created.
- When a class’ lifeline has a cross on it, it is ‘deleted’.



Communication diagrams

- A communication diagram does the same as a sequence diagram, except it uses numbers instead of graphical axes.
- You start from the user and work your way across.
- The first interaction is marked with ‘1’. The interactions born from that interaction are labelled ‘1.1’, ‘1.2’, ‘1.3’ etc.

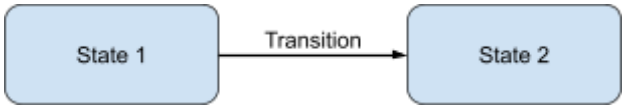
- Once there are no more interactions marked '1.x', you move onto the second interaction marked '2' and '2.x' etc.
- Communication diagrams are more compact than sequence diagrams, but it can be harder to follow.



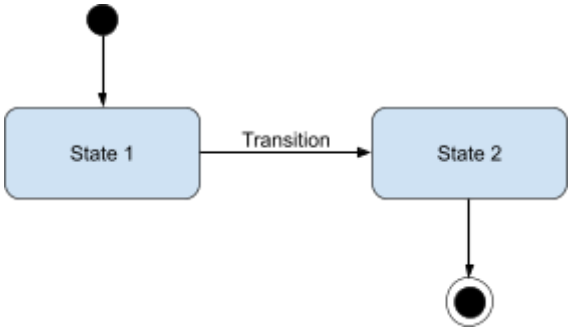
State machine diagrams

- A state machine diagram shows how a system goes through one state to another in response to external events.
- A **state** is a condition satisfied by the attributes of an object. The system can only be in one state at a time. It is represented by a rounded rectangle.

- A **transition** is a change of state triggered by an event (press button), condition (sum > 10) or time (after(1s) *meaning after 1 second*). It is represented by an arrow connecting one state to another.

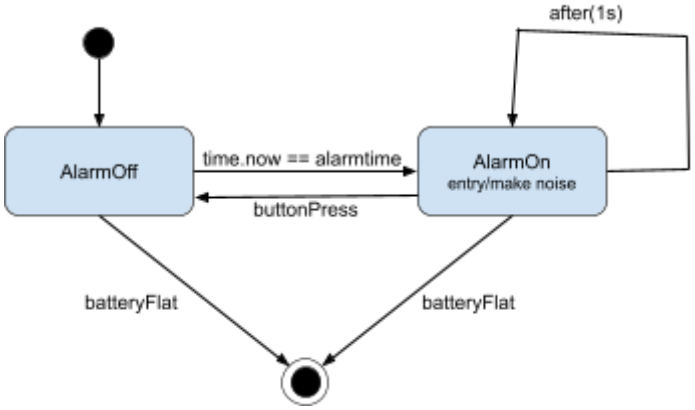


- When a system is turned **on**/**off**, that is represented by a thick dot and a thick dot with a border around it, respectively.

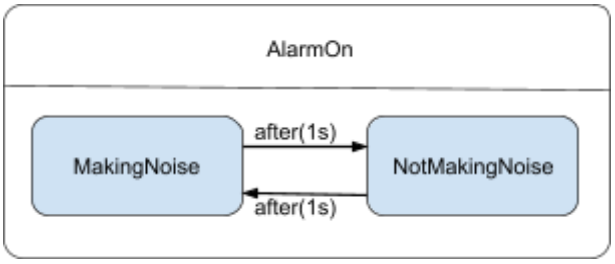


- An **internal transition** is a transition that does not change its state. It's represented with an arrow going from a state and ending back at the same state.
- An **activity** is an action performed by the system in a specific state, like 'count ticks' or 'make noise'. It is accompanied by a **label**, which is a condition that must be true

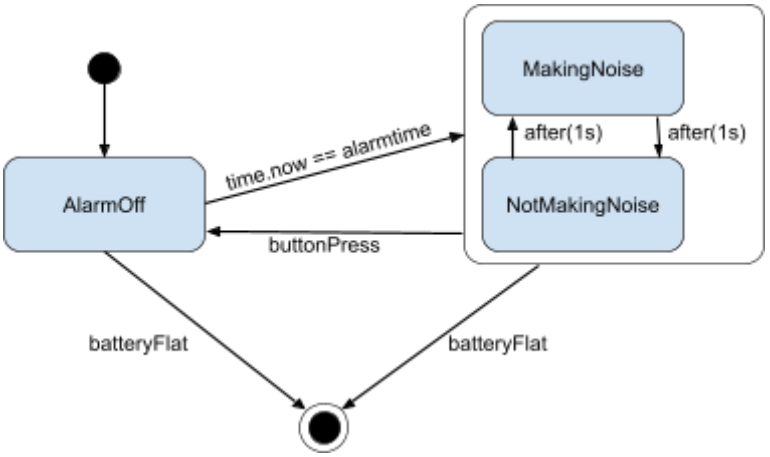
for the activity to take place, like ‘entry’ which means ‘when this state has been entered’, or ‘exit’ which means ‘when this state has been exited’, or ‘do’ which means ‘execute the activity as long as we’re in this state’.



- A **nested state machine** is a special kind of state machine that represents a state from a higher-level state machine. You can use these instead of internal transitions if things get too complex for them. Every state could be represented as a nested state machine.



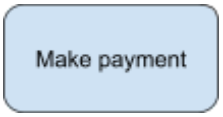
- A **substate** is like a nested state machine, but it’s a part of its parent state machine.



Activity diagrams

- An activity diagram is a bit like a state machine diagram, except each transition is caused by internal events.
- Think of it like an advanced flow chart.
- It differs from a state machine diagram by focusing on ‘control’, ‘behaviour’ and ‘dataflow’ as opposed to the state (set of attributes) of single abstractions (objects).
- One of the best things about activity diagrams is that it helps to visualise concurrent activities (e.g. separate threads).

- There are three kinds of nodes:
    - **Control** nodes - defines a control/deviation in where the activity goes (special icons)
    - **Executable** nodes - defines an action to be performed (e.g. make payment)



    - **Object** nodes - defines a model (e.g. document)

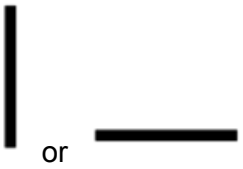
- Imagine one of the system's threads, traversing across the diagram, going from node to node. That is called a **'token'**, and they are created at an initial node and ends at a final node.

- An **initial node** is where a token starts. It is represented with a filled dot.

- A **final node** is where a token ends. There are two kinds of final nodes:
    - An **activity final node** marks the end of the activity. It is represented with a filled dot with a border. The 'goal' of the activity has been accomplished.

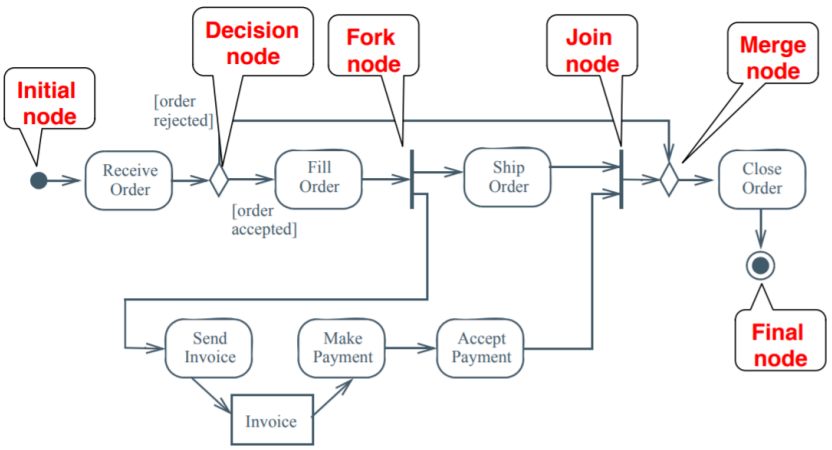
    - A **flow final node** marks the end of a token, but other tokens can still traverse through the activity. The 'goal' of the activity has not yet been accomplished.

- A **fork/join node** is a node that either splits up or joins up tokens respectively. This either creates or stops concurrent processes. For a join node's output flow to begin, it must first receive all possible input tokens.

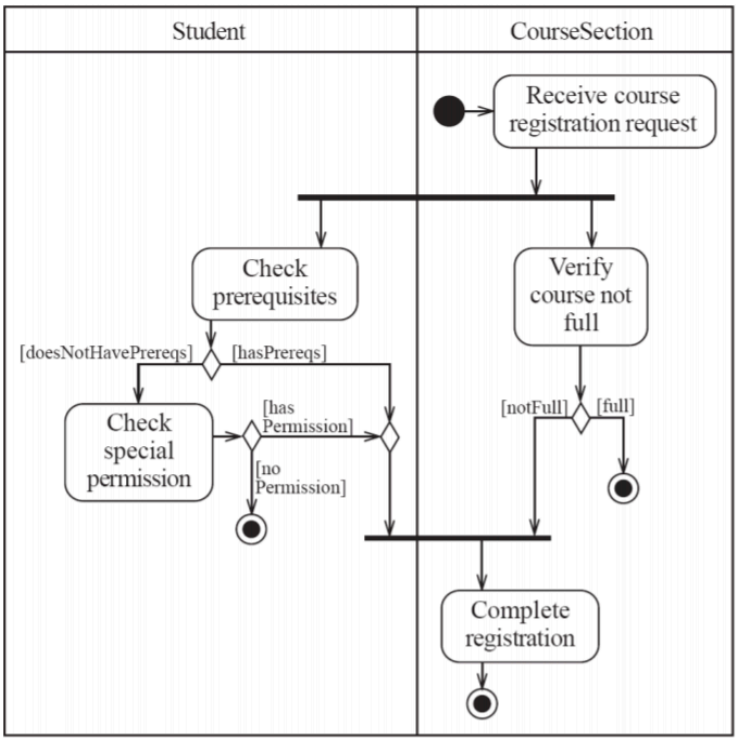


- A **merge/decision node** is a node that can either merge tokens together or decide to direct tokens one way or another given a condition. A merge node's output flow can begin as long as at least one input token is passed through.

- Example:



- A **swimlane** is a partition of the diagram that belongs to a class.
- All activities done in a swimlane are being done by their respective class.
- Example:



Model validation and verification

- **Verification** is an equivalence check between the transformation of two models
- **Validation** is the comparison of the model with reality (requirements should be validated with the client and the user)
- A good way to tell the difference is through Boehm's expression:
    - **Validation**: Are we building the right product?
    - **Verification**: Are we building the product right?

- Requirements validation involves several checks:

    - Correctness (represents client's view of the system?)
    - Completeness (every scenario is described?)
    - Consistency (no components contradict each other?)
    - Ambiguity (model describes one system, not many?)
    - Realistic (model can be implemented?)
- You should also check for:
    - Syntax
    - Consistent naming
    - No dangling associations (points to nowhere)
    - Missing classes (mentioned but not defined anywhere)

Event-B

Introduction to Event-B

The basics

- Event-B is a method of software development based off of B-method, and is considered an 'evolution' of B-method.
- Event-B notation uses set theory and predicate logic for describing software.
- We can describe systems and what they can do, for example:
    - Let's just say we have a system that monitors people entering and leaving the building.
    - We can model that like this in Event-B:

```
context BuildingContext
sets USER
end

machine Building
variables register in out
invariants

inv1: register ⊆ USER // set of registered users
inv2: register = in ∪ out // all registered users must be
                          // either inside or outside
inv3: in ∩ out = {} // no user can be inside and outside

initialisation in, out, register := {}, {}, {}
events
 Enter                                Leave
   any s where                          any s where
     s ∈ out ∩ register                   s ∈ in ∩ register
   then                                 then
     in := in ∪ {s}                       in := in \ {s}
     out := out \ {s}                     out := out ∪ {s}
   end                                  end
```

- Let's go over the keywords for this:

Keyword	Example	Explanation
context	`context BuildingContext`	Defines the scope of the machines/systems,

		e.g. everything we describe is within the context of the building, namely, “BuildingContext”.
sets	`sets USER`	Defines the types of groups within the whole system, e.g. there is a “USER” group within the building’s context.
end	`end`	Simply marks the end of a clause, for example the ‘end’ in ‘context ... sets ... end’ marks the end of the clause describing the context of the system. It is also used with events.
machine	`machine Building`	Marks the start of a clause that describes a machine within the context of the system, e.g. there is a “Building” machine in the building’s context.
variables	`variables register in out`	Defines the variables within a machine, e.g. ‘register’, ‘in’ and ‘out’ are variables within the ‘Building’ machine.
invariants	`inv1: register ⊆ USER`	Invariants are rules that the variables must abide by, e.g. the ‘register’ variable must always be a subset of ‘USER’.
initialisation	`initialisation in, out, register := {}, {}, {}`	Defines variables at the start of the machine’s lifetime, e.g. ‘in’, ‘out’ and ‘register’ variables will all be empty sets at the start
events	`events Enter ... Leave ...`	Marks the start of the definition of events related to a machine, e.g. ‘Enter’ and ‘Leave’ are events defined in the machine ‘Building’.

Defining types

- You can define the types of variables in two ways like so:

$$w \in P(T) \qquad\qquad w \subseteq T$$

- With ‘w’ being the set of variables of a certain type ‘T’.
- Example:

$$\{3, 4, 5\} \in P(Z) \qquad\qquad \{3, 4, 5\} \subseteq Z$$

- Here, we define that the elements of the set {3, 4, 5} are all integers.

Relations and functions

Ordered pair

- **An ordered pair is a tuple with two elements**.
- It can represent a domain element being mapped to a range element: (x, y)
- They are usually written like this:

- $x \mapsto y$

Cartesian product

- The **cartesian product** gives you **every possible ordered pair between two sets**.
  - For example, $x \mapsto y \in X \times Y$
  - Given that x is within X and y is within Y
- Therefore, you can use the cartesian product to express the 'typing' of an ordered pair.
  - For example, $2 \mapsto 4 \in Z \times Z$
  - Where 'Z' is the set of all integers.

Relations

- **A relation is a set of ordered pairs**.
- They allow mapping from a set of elements to another, without the restrictions of a function.
- The notation for a relation is this:
  - $T \leftrightarrow S = \mathbb{P}(T \times S)$
- So when you see the symbol '↔', just think "<u>a set of all possible relations/mappings between these two sets</u>"

Domain and range;

- The **domain** is the **set of all the elements of the first part of a relation**.
- The **range** is the **set of all the elements of the second part of a relation**.
- To specify the domain of a relation, you write this:
  - $dom(R)$
- To specify the range of a relation, you write this:
  - $ran(R)$

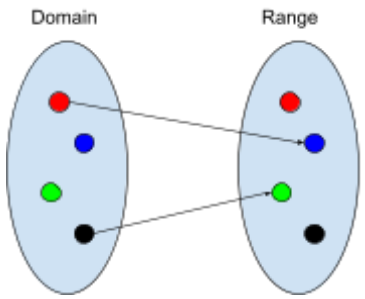
Relational image

- 'Relational image' simply refers to mapping input elements through a relation and outputting a set of output results.
- It is written like this:
  - $R[A]$
  - Where 'R' is the relation and 'A' is a set of input elements.
- Think of it as Event-B's version of a function/relation call, except you can input a set of input elements and receive a set of output elements.

Partial function

- A partial function is like a normal function, except not all of the domain elements have to map to an element of the range.
- A partial function is defined like this:

    - $f \in x \nrightarrow y$
- So when you see an arrow with a line through it, think "set of functions where not all the domain elements map to the range elements".



Well-definedness

- A function is **well-defined** if **for each x there is a unique y such that f(x) = y**.
- Well-definedness is one of the criterion for a function: no well-definedness, no function.
- An example of a function that is not well-defined is the following:
    - f(x) = the first digit after the decimal point
    - so therefore f(1) = 0 because 1.**0**000
    - but also f(0.9999...) = 9 because 0.**9**999...
    - But 0.999.... and 1 are the same, so that means the value '1' has two outputs, which makes f(x) not well-defined.

Restrictions and subtractions

- You can restrict or subtract domains and ranges. This means either filter out domains/ranges to leave the ones you want, or getting rid of domains/ranges that you don't want.

Type of restriction or subtraction	**Syntax**	**Explanation**	**Example**
Domain restriction	A ◁ R	Filter out domain elements, leaving only the domain elements specified.	{1,3} ◁ {1→10,2→20,3→30} = {1→10,3→30}
Domain subtraction	A ⩤ R	Remove specified domain elements.	{1,3} ⩤ {1→10,2→20,3→30} = {2→20}
Range restriction	R ▷ B	Filter out range elements, leaving only the range elements specified.	{1→10,2→20,3→30} ▷ {10,30} = {1→10,3→30}
Range subtraction	R ⩥ B	Remove specified	{1→10,2→20,3→30}

		range elements.	⩥ {10,30} = {2→20}

- By using these, you can filter out or remove certain domain / range values.

Function overriding

- You can override a function f with function g by using this notation:
  - f ⊲ g
- By doing this, you're overwriting / adding new mappings into a function.
- Example:
  - { 1 → 10, 2 → 20 } ⊲ { 3 → 30, 1 → 50 } = { 1 → 50, 2 → 20, 3 → 30 }

Total function

- A **total function** is a **well-defined partial function**.
- It's what you'd traditionally know as a function.
- It's represented like this:
  - f ∈ X → Y

Relational inverse

- A relational inverse is the same as an inverse function; it swaps the domain and the range.
- If you have a relation R ∈ S ↔ T, then an inverse relation is R^{-1}.
- This works for functions too, as long as it's <u>injective</u> *(if it's not surjective, then the function inverse will be a relation / partial function)*.

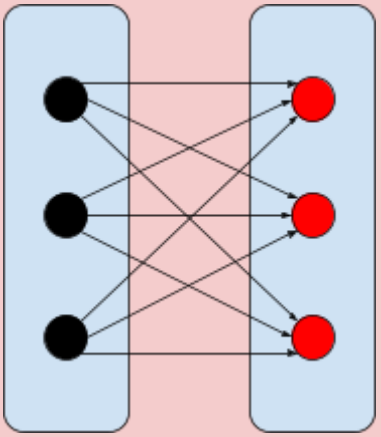
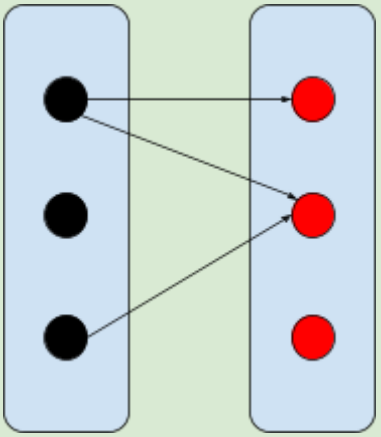
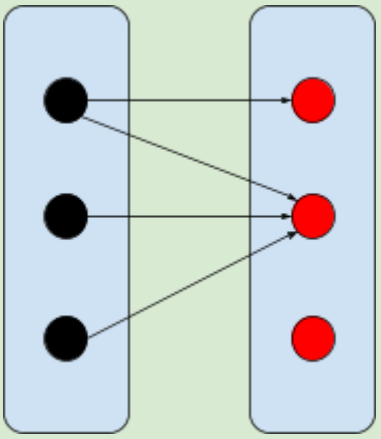
Relational composition

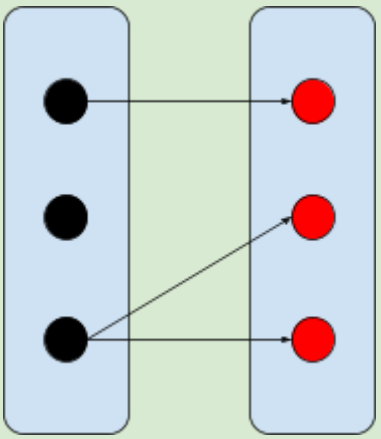
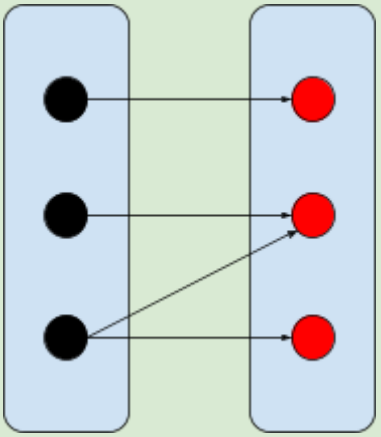
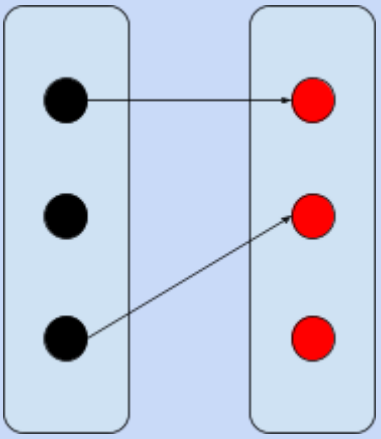
- Like function composition, relational composition is the process of performing multiple relations, one after another, on an input.
- If you had relations Q ∈ S ↔ T and R ∈ T ↔ U, you could create a composition of these by writing Q ; R, where Q ; R ∈ S ↔ U.

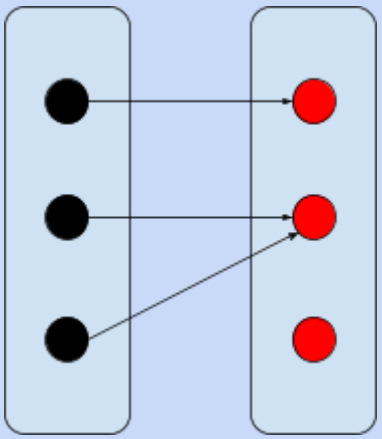
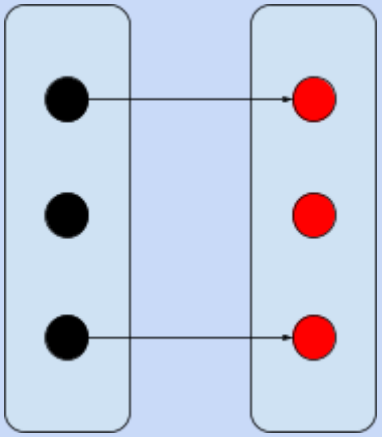
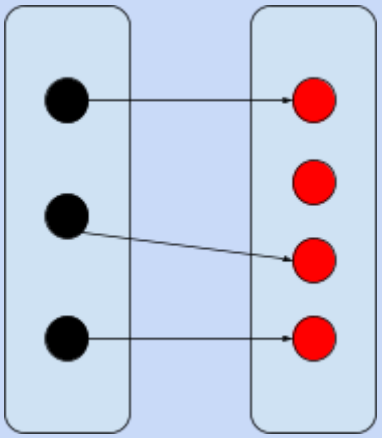
Symbols

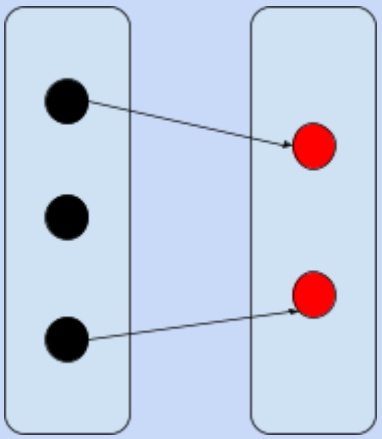
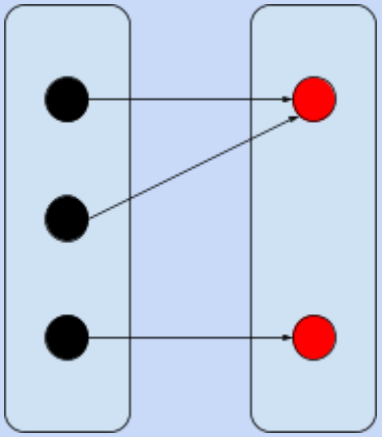
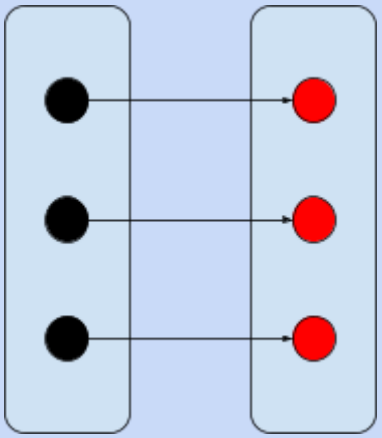
- All the symbols for relations/functions can be found here:

Symbol	Meaning	Example / Illustration
↦	Ordered Pair Not a relation or a function, just one pairing from one element to another.	$x \mapsto y$

$\times$	Cartesian product Every single possible pairing from elements of one set to another.	
Relations **These kinds of mappings can map from one domain element to multiple range elements.**		
$\leftrightarrow$	Normal relation A relation of any mapping from one set to another. Think of this as simply a subset of the cartesian product.	
$\twoheadleftarrow\!\!\rightarrow$	Total relation Like a normal relation, but <u>all of the domain</u> is mapped to something.	

$\leftrightarrow\!\!\!\!\twoheadrightarrow$	**Surjective relation** Like a normal relation, but <u>all of the range</u> is being mapped by something.	
$\twoheadleftarrow\!\!\!\!\twoheadrightarrow$	**Total surjective relation** Like a normal relation, but all elements from the domain map to all elements in the range.	
Functions **Like relations, but you cannot map one domain element to multiple range elements.**		
$\nrightarrow$	**Partial function** Not all of the domain has to be mapped.	

$\rightarrow$	**Total function** Maps all elements of the domain to an element of the range.	
$\rightarrowtail\!\!\!\!\!+$	**Partial injection** Not all elements of the domain have to be mapped, but the ones that are need to map to unique range elements.	
$\rightarrowtail$	**Total injection** All elements from the domain need to be mapped, and they all need unique range elements.	

$\nrightarrow\!\!\!\!\rightarrow$	**Partial surjection** Not all domain elements need to be mapped, but all elements from the range need to be mapped from something.	
$\twoheadrightarrow$	**Total surjection** All elements from the domain need to be mapped, and all elements from the range need to be mapped by something.	
$\rightarrowtail\!\!\!\!\twoheadrightarrow$	**Bijection** All elements from the domain need to be mapped, all the elements from the range need to be mapped and no two domain elements should map to the same range element.	

Multiplicity to association

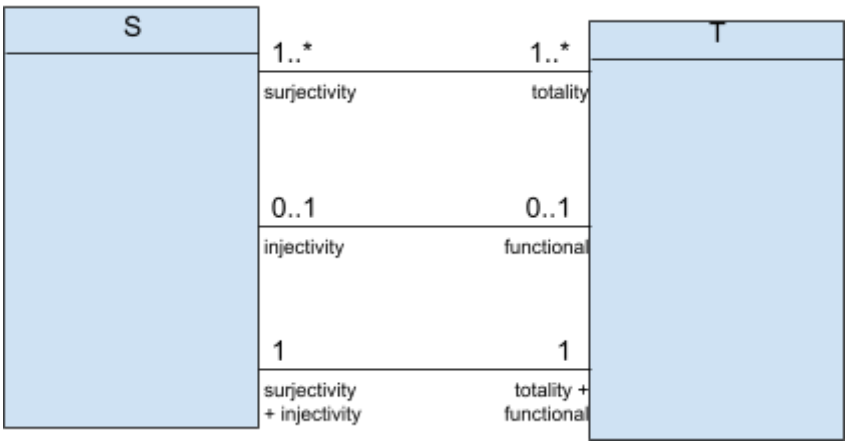
- Given the multiplicity [x] ---- [y], where [x] is a multiplicity like 0..1 or *, and the same with [y], then the following table can be used to look up the type of association with this multiplicity:

	0..1 Injective	1 Injective + surjective	1..* Surjective	* None

0..1 Partial function (functional)	Partial injective function	Partial injective + surjective function	Partial surjective function	Partial function
1 Total function (totality + functional)	Total injective function	Total injective + surjective function (bijection)	Total surjective function	Total function
1..* Total relation (totality)	Total injective relation	Total injective + surjective relation	Total surjective relation	Total relation
* Partial relation (none)	Partial injective relation	Partial injective + surjective relation	Partial surjective relation	Partial relation

- Like how '*' means 'none', '0..n' also means 'none', and 'n..m' is equivalent to '1..*'.
- If an association is not 'total', it is 'partial' and vice versa.
- If an association is not 'functional' (a function), it is a relation, and vice versa.
- For associations like "partial injective + surjective relation", there's no such symbol that represents that. When that happens, you can take the inverse and model that, which would be "total function", which is represented with '→'.

- Additionally, you can remember this diagram, with class S on the left and class T on the right:



- For more information on this, read this.

Classes and associations

- In Java, when you create a class, you put its attributes inside the class file, where the class is declared and defined.
- However, in Event-B, you create a *base class* and you add attributes by initialising relations whose *domain is the class* and the *range is the attribute*.
- These attributes can be referenced through relations.

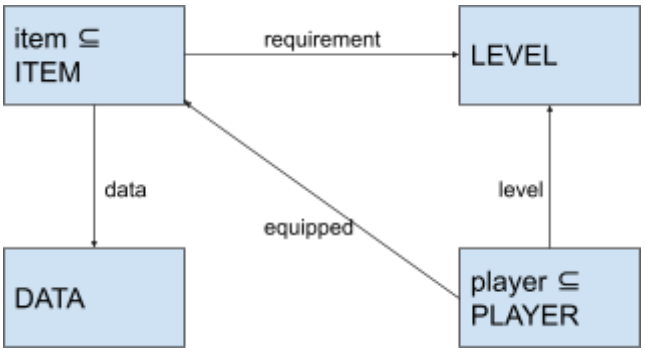
- So, effectively, the attributes of a class and the class itself are defined separately in Event-B, but can still logically be thought of as “one belonging to the other”.

Primary and secondary carrier sets

- In Event-B, there are two types of carrier sets: primary and secondary.
- Primary:
    - These are the objects that actually exist; Event-B handles management of these (creation, modification, deletion etc.)
- Secondary:
    - These are attributes for the primary carrier sets. You can’t actually initialise them; they must belong to an object.

Class diagrams

- In a class diagram, a good convention is to make object names ‘lowercase’ and class names ‘UPPERCASE’.
- Let’s create an example. Let’s say you have a character in an MMORPG, and you are a certain level. There are items you want to equip that have a ‘level requirement’, and you must be a certain level to equip it. Each item also has a description, which we will call ‘DATA’.
- We can model this in a class diagram first before jumping head-first into Event-B:



- From this class diagram, we can see that ‘player’ and ‘item’ are set objects containing all the ‘PLAYER’s and ‘ITEM’s, respectively.
- ‘ITEM’ and ‘PLAYER’ are primary carrier sets so we introduce set objects for them. ‘DATA’ and ‘LEVEL’ are secondary carrier sets, so we do not.
- An ‘ITEM’ has two attributes: the data (description) and its level requirement.
- A ‘PLAYER’ has two attributes: their level, and the items they’ve equipped (for this model, a player can only equip one item. This is to keep the ‘equipped’ relation simple).
- Another interesting thing is that all items must have a description and a requirement, and all players must have a level. However, not all players must have an equipped item, which means the ‘equipped’ relation must be a partial function (not all players map to an item).

Class declaration in Event-B

- Now that we have a general idea of what to do, we can start modelling this system in Event-B:

```
sets ITEM DATA PLAYER
constants LEVEL
axioms LEVEL = 1..99
variables item, player, data, requirement, level, equipped
invariants
  item ⊆ ITEM
  player ⊆ PLAYER
  data ∈ item → DATA
  requirement ∈ item → LEVEL
  level ∈ player → LEVEL
  equipped ∈ player ⇸ item
initialisation
item := {} player := {} data := {} requirement := {} level := {}
```

- In the code above, we have created 3 types: ‘ITEM’, ‘DATA’ and ‘PLAYER’. Note that ‘LEVEL’ is a constant, and we have set an axiom for ‘LEVEL’ to be between 1 to 99 (1..99 means 1 to 99 inclusive). This adds validation to ‘LEVEL’, ensuring that it doesn’t reach strange values (like negative).
- The attributes of ‘ITEM’ and ‘PLAYER’ are stored as ‘data’, ‘requirement’, ‘level’ and ‘equipped’. They are all functions that map items/players to their respective attributes.

Class definition in Event-B

- So now we have a good base in Event-B, but we need to define events so that things can actually happen.

AddPlayer

- When a player is created, they must have a level. Usually, a player starts at level 1, but for this example, a player can start at any level. Therefore, we need a level parameter as well as a player parameter.
- Additionally, when a player is created, they don’t have any items equipped, so that parameter isn’t needed.

```
AddPlayer ≙
  any p, l where
    p ∈ PLAYER
    p ∉ player
    l ∈ LEVEL
  then
    player := player ∪ {p}
    level(p) := l
```

```
  end
```

AddItem

- When an item is created, they must have a requirement level and a description. Therefore, we need three parameters: the item, the requirement level and the data.

```
AddItem ≙
  any i, l, d where
    i ∈ ITEM
    i ∉ item
    d ∈ DATA
    l ∈ LEVEL
  then
    item := item ∪ {i}
    data(i) := d
    requirement(i) := l
  end
```

ReadItem

- When a player reads an item, they're trying to get its description. However, a player can only see its description if it meets its level requirement (unconventional for an MMORPG, but we can still model it).
- In Event-B, there is no 'return' statement. Instead, you have a return parameter.
- Therefore, we'll need three parameters: the item, the player and the result.
- Notice that the 'result' parameter is being set with the conditions in the 'where' clause. The reason why we don't do this in the 'then' clause is because nothing is being explicitly changed/set.

```
ReadItem ≙
  any i, p, result where
    i ∈ item
    p ∈ player
    level(p) ≥ requirement(i)
    result = data(i)
  end
```

WriteItem

- When a player writes to an item, they change its description as long as they meet its level requirement (again, unconventional for an MMORPG, but it makes good Event-B examples).
- WriteItem requires the item to change, the player to make the change and the data to set the item to.

```
WriteItem ≙
  any i, p, d where
    i ∈ item
```

```
    p ∈ player
    d ∈ DATA
    level(p) ≥ requirement(i)
  then
    data(i) := d
  end
```

ChangeRequirement

- Perhaps an admin wants to change the level requirement of an item, or maybe the player upgrades the item and its requirement increases.
- When changing the item requirement, we need to make sure that the item's requirement doesn't exceed the level of the player who has this item equipped.
- We need two parameters: the item and the level to change it to.
- Since only one item can be equipped by one player, we can use the inverse function 'equipped^{-1}'.

```
ChangeRequirement ≙
  any i, l where
    i ∈ item
    l ∈ LEVEL
    equipped⁻¹;level(i) ≥ l
  then
    requirement(i) := l
  end
```

LevelUp

- Levelling up is simple; we just need to make sure the player doesn't exceed the maximum level.
- To check if the player's level won't exceed the maximum, we could check if the level of the player plus one is less than 99. However, what if the maximum level changes?
- If the level value goes over the maximum, it will stop being of type 'LEVEL' (because the definition of being of type 'LEVEL' means being within the appropriate range), therefore we can check if it exceeds the maximum level by checking if the level plus one is still of type 'LEVEL'. If it is, then the player's level is still within the correct range after levelling up.
- We only need one parameter: the player.

```
LevelUp ≙
  any p where
    p ∈ player
    level(p) + 1 ∈ LEVEL
  then
    level(p) := level(p) + 1
  end
```

EquipItem

- When a player equips an item, they must meet its level requirement.
- We need two parameters: the player and the item.

```
EquipItem ≙
  any p, i where
    p ∈ player
    i ∈ item
    level(p) ≥ requirement(i)
  then
    equipped(p) := i
  end
```

RemovePlayer

- Let's just say some player hacked the game and needs to be banned (removed) by an admin.
- When we remove the player, we need to check the functions going to and from 'player'. It wouldn't make sense for an item to be equipped by a player that doesn't exist.
- The relations that interact with players are 'equipped' and 'level'.
- For this event, we only need one parameter: the player to be removed.

```
RemovePlayer ≙
  any p where
    p ∈ player
  then
    player := player \ {p}
    equipped := {p} ⩤ equipped
    level := {p} ⩤ level
  end
```

RemoveItem

- Let's just say an item was deemed too OP and needs to be removed in-game by an admin.
- When we remove the item, we need to check the relations going to and from the items, like in 'RemovePlayer'.
- The relations that interact with the items are 'requirement', 'data' and 'equipped'.
- For this event, we only need one parameter: the item to be removed.

```
RemoveItem ≙
  any i where
    i ∈ item
  then
    item := item \ {i}
    requirement := {i} ⩤ requirement
    data := {i} ⩤ data
```

```
    equipped := equipped ⩤ {i}
  end
```

Extension refinement

- Refinement is the process of:
  - improving the functionality being modelled, or
  - explain how some purpose is achieved

Refinement step

- A refinement step is when we remodel a model M1 into M2.
  - M1 is an abstraction of M2
  - M2 is a refinement of M1
- We can perform lots of refinement steps to go from M1 to M2 to M3 to M4 etc.

What can you do with refinement

- With extension refinement, you can:
  - *Add additional variables and invariants*
  - *Extend existing events* to act on additional variables
  - *Add new events* to act on additional variables
- So you can think about refinement in the same way that classes inherit each other in OOP languages.

Syntax

Machine extension

- To show that machine M2 refines another machine M1, you write:

```
machine M2
refines M1
variables ...
invariants ...
events...
```

- When you do this, all the invariants are inherited from M1, however, you need to state the variables you need from M1 again in M2.

Event extension

- To extend events, you can write:

```
E2 extends E1 ≙
  any <only additional parameters> where
    <additional conditions>
  then
    <additional actions>
```

```
end
```

- Where E1 is an event in M1 and E2 is a child event of E1 in M2.
- If they both have the same name, it's fine to write "event extends event", and the assumption will occur where there is a parent event of the same name in the parent machine.

Examples

- Given the machine:

```
context BuildingContext
sets USER
end

machine Building1
variables register in out
invariants

inv1: register ⊆ USER // set of registered users
inv2: register = in ∪ out // all registered users must be
                          // either inside or outside
inv3: in ∩ out = {} // no user can be inside and outside

initialisation in, out, register := {}, {}, {}
events
```

```
Enter
  any s where
    s ∈ out ∩ register
  then
    in := in ∪ {s}
    out := out \ {s}
  end
```

```
Leave
  any s where
    s ∈ in ∩ register
  then
    in := in \ {s}
    out := out ∪ {s}
  end
```

- We have a building, and people can come in or come out. We store who is in the building and who is outside the building.
- What if we wanted to model someone going into a certain room, say, room 2B?
- We could use extension refinement to extend this building to model this:

```
machine Building2
refines Building1
variables register in out room2B
invariants

inv1: room2B ⊆ in // Everyone in 2B must be in the building
```

```
initialisation in, out, register, room2B := {}, {}, {}
events
 Enter2B ≙
   any s where
     s ∈ in ∩ register
   then
     room2B := room2B ∪ {s}
   end

 Leave2B ≙
   any s where
     s ∈ room2B
   then
     room2B := room2B \ {s}
   end
```

Queues

- In Event-B, to add a queue, you have to implement one yourself.
- <u>To create a queue, you need injective functions</u>.

Injective functions

- An **injective function**, also known as a **one-to-one function**, is a function where each element in the domain maps to one unique element in the range.
- For example, if you have an injective function f(x), you cannot have f(x1) = y and f(x2) = y as well, for x1 != x2.
- An injective function can be represented like this:
  - $f \in X \rightarrowtail\!\!\!\!\rightarrow Y$
- A total injective function is an injective function where every element of the domain maps to an element of the range. It is represented like this:
  - $f \in X \rightarrowtail Y$

Single queue

- On a queue, elements have a position. In queues in Event-B, the smallest position element will be dequeued next, and the biggest position element was the most recently added element.
- Therefore, we can model a queue like this:

- We can model this with these invariants:

```
inv1: element ⊆ ELEMENT
inv2: position ∈ element ⤖ POSITION
```

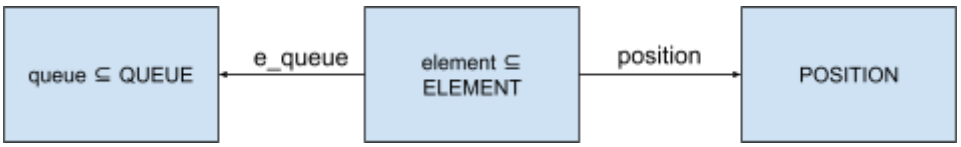
- Now, we can add events to enqueue and dequeue elements:

```
event EnqueueJob ≙
 any e p
 where
  e ∉ element
  p ∈ POSITION
  ∀k· k ∈ element ⇒ p > position(k) // p is the highest position of
all elements in the queue
 then
  element ≔ element U {e}
  position(e) ≔ p
end
```

```
event DequeueJob ≙
 any e
 where
  e ∈ element
  ∀k· k ∈ element ⇒ position(e) ≤ position(k) // e is the lowest
positioned element of the whole queue
 then
  element ≔ element \ {e}
  position ≔ {e} ⩤ position
end
```

Multiple queues

- To handle multiple queues, each element must be linked to one queue. Elements can still have positions, but two elements in the same queue cannot have the same position.



- An invariant for this would be:

```
inv1: e_queue ∈ element → queue
```

- Before, we had an invariant that linked elements to a single position.
- We need to add something extra to make sure no two positions on the same queue is the same for two elements:

```
inv2: ∀e,k·e ∈ element ∧ k ∈ element ∧
   e ≠ k ∧
   e_queue(e) = e_queue(k)
⇒  position(e) ≠ position(k)
```

- Now we can add the events:

```
event QueueJob ≙
```

```
 any e p q
 where
  e ∉ element
  p ∈ POSITION
  q ∈ queue
  ∀k· k ∈ element ∧ e_queue(k) = q ⇒ p > position(k) // p is the
highest position of all elements in the queue
 then
  element ≔ element ∪ {e}
  position(e) ≔ p
  e_queue(e) ≔ q
end


event DequeueJob ≙
 any e, q
 where
  e ∈ element
  q ∈ e_queue(e)
  ∀k· k ∈ element ∧ e_queue(k) ⇒ q > position(e) ≤ position(k) // e
is the lowest positioned element of the whole queue
 then
  element ≔ element ∖ {e}
  position ≔ {e} ⩤ position
  e_queue ≔ {e} ⩤ e_queue
end
```

Modelling state machines

What is a state machine

- A state machine is a model that has distinct states depending on what they're doing.
- There are also transitions from one state to another.
- For example, a state machine for a switch could be ON and OFF, and a transition for that could be toggling (pressing) the switch.

Modelling states

- You use *sets* to model states:

```
in ⊆ user \\ users in the 'in' state
out ⊆ user \\ users in the 'out' state
```

- Then you would use events to represent the transitions between states:

```
Enter ≙
  any u where
    u ∈ out
```

```
    u ∉ in
  then
    in := in U {u}
    out := out \ {u}
  end


Leave ≙
  any u where
    u ∈ in
    u ∉ out
  then
    in := in \ {u}
    out := out U {u}
  end
```

Invariants relating to the states

- To make sure you cannot be in two states, add invariant:

in ∩ out = ∅

- To make sure you cannot have the same element in two states.

Partition operator

- The partition operator allows you to partition a set without making all the invariants and sets yourself. Think of it like syntactic sugar.
- The syntax goes like this:
  - partition(S, T_1, ... T_n)
  - Where:
    - S is the set with all the elements
    - T_1 up to T_n are all the partitions
- In other words:
  - $S = T_1 \cup ... \cup T_n$
  - $T_i \cap T_k = \emptyset$ each i,k where i != k
- You can use this to create state machines in Event-B easily.

- Example using the in-out state machine like above:
- partition(user, in, out)

Set comprehension

- The set of elements x that satisfy a predicate:
- $\{ x \mid x \in X \}$
- So, for example, the set of players whose levels are over level 10:
- $\{ p \mid p \in \text{players} \wedge \text{level}(p) > 10 \}$
- It's a way of getting a set whose elements satisfy a predicate.

Query event

- A **query event** is an **event that fetches data from the model and returns a result**. It doesn't affect the model.
- For example, getting the set of friends of a user:

```
GetFriends ≜
  any u, result where
    u ∈ users
    result = { f | f ∈ users ∧ f ∈ friends[{u}] }
  end
```

Miscellaneous

Test design

Testing terms

- **Error**: When a person makes a mistake and the system does not perform the way you intended it to.
- **Fault**: The actual part of the system that causes it to fail, also known as a "defect" or a "bug".
- **Failure**: When the system does not function the way it should due to a fault.

- To remember this, think of this sentence: "You make an error while writing the program, leaving a fault in the code and therefore it results in a failure".
- Not all faults lead to failures. This is because some use cases do not encounter the conditions for any bugs, so the system works fine for that use case.

Test cases

- We test by giving a program a bunch of inputs, and then expecting certain outputs. If the outputs are what we expected, the test cases pass. If not, they fail.
- Test cases can be laid out like this:

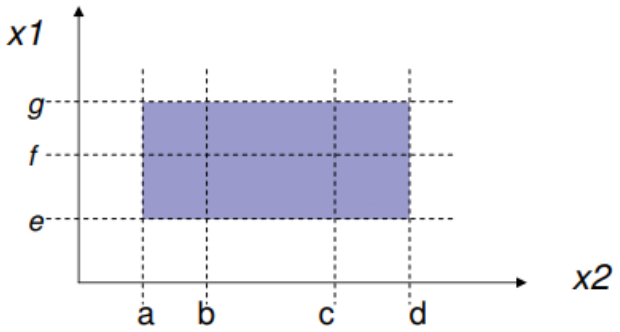
ID, date	Module / unit name	Test specification	Expected result	Pass?

- You could also add "Actual output" and "Observations" on a combined test plan & report.
- You need positive test cases and negative test cases.
    - Positive: If the program does this as expected, test succeeds
    - Negative: If the program fails here / throws exception, test fails

Equivalence class testing

- Equivalence class testing is testing using equivalence classes.
- Equivalence classes are sets where all of the elements in that set are equivalent to each other in some way.
- For example, let's say you had a game where if your health is 3 points or lower, you're on "low health".
- Therefore, there are two equivalence classes: low health and normal health, and the inputs "2 points" and "3 points" are in the same equivalence class, because they're categorised the same in the given condition.
- The whole point of equivalence class testing is to test each value from each equivalence class, so we can ensure:
    - Completeness
    - Non-redundancy
    - Minimise testing effort

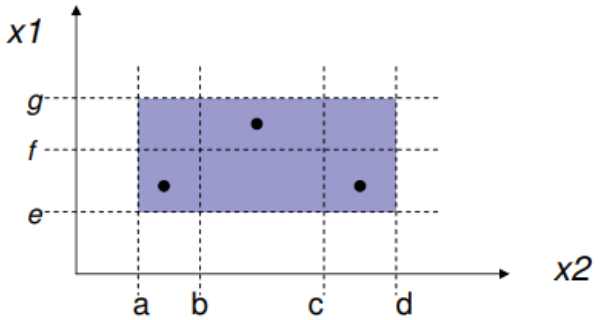
- Equivalence classes can be put on a graph for a visual representation. For example, for a program with two inputs, you could have this graph for equivalence classes:



- The areas between the dotted lines (shaded in parts) represent equivalence classes, and the letters 'a', 'b', 'c', 'd' etc. all represent the bounds of the ranges.

Weak normal ECT

- Weak normal ECT relies on the "single fault assumption", which states that failures are rarely the result of two other faults simultaneously occurring.
- Therefore, we only need one test per equivalence class:

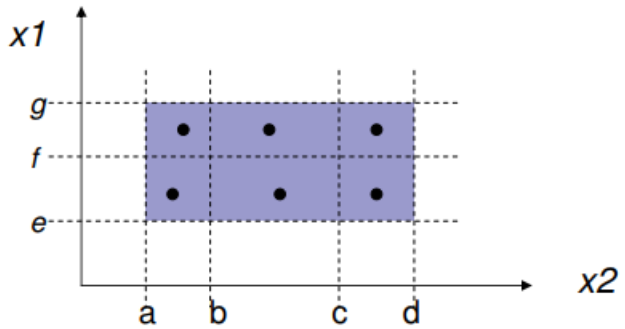


- Notice that for each equivalence class, there's a minimum of one test case.

- To find the number of test cases in weak normal ECT, find the equivalence relation with the biggest number of equivalence classes, and then pick that number, for example in the picture above x1 has 2 ECs, x2 has 3 ECs, so we need 3 test cases.

Strong normal ECT

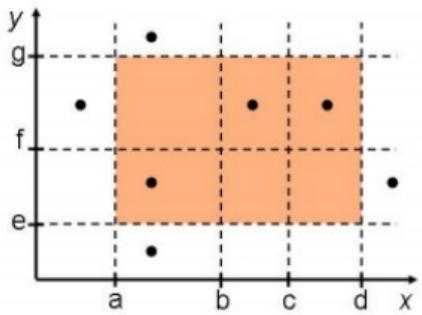
- Strong normal ECT relies on the assumption that one error may cause lots of faults.
- Therefore, we need at least one test case on every possible equivalence class relation.
- More formally, there is a test case for each element of the cartesian product of the equivalence relation:



- Notice how for every block made up of two equivalence classes, there is a test case. If there were three variables, you'd need a test case for every cuboid made up of three equivalence classes.
- To get the number of test cases, multiply the cardinality of all the equivalence classes with each other, for example there are 2 ECs in x1, 3 ECs in x2, 2 * 3 is 6, so we need 6 test cases.

Weak robust ECT

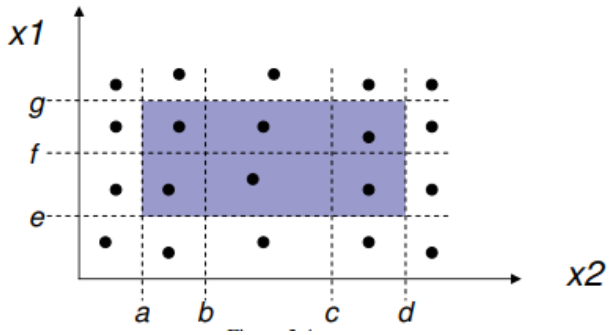
- Weak robust ECT is the same as weak normal ECT, except we have invalid values, too.
- For each variable, we have at least one invalid value on either sides of the valid ECs; all the rest are valid:



- Notice that for variables x and y, both invalid ECs on either side have test cases.
- The number of test cases is:
  - $\max_i\{S_i\} + \sum c_i$
  - Where c_i is the number of out-of-range cases for a variable i. Usually, it's 2.

Strong robust ECT

- Strong robust ECT is the same as strong normal ECT except we have invalid values, too; much like how weak robust ECT is to weak normal ECT.
- We basically fill in every possible block with a test case.
- More formally, we produce test cases for all valid and invalid elements of the cartesian product of all the equivalence classes:



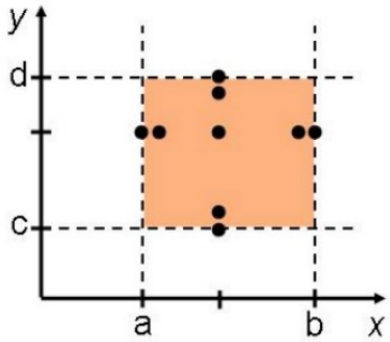
- Notice how every space is occupied by a test case, even the invalid ones.
- The number of test cases in this is:
  - $\Pi\{S_i + c_i\}$
  - Which means the product of all $S_i + c_i$.

Boundary value testing

- Instead of the test cases being right in the middle of an equivalence class, boundary value testing tests the boundaries of the equivalence classes.

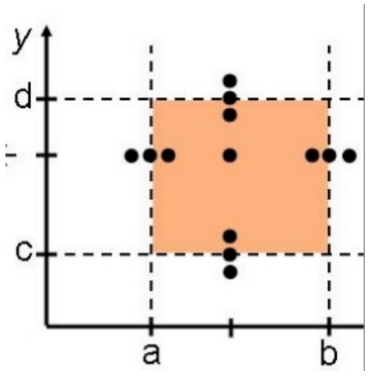
Normal BVT

- With normal BVT, you pick all values on the boundaries, values right next to the boundaries and a value in the middle of the range.



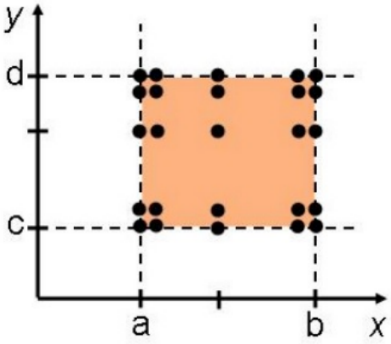
Adding robustness

- Robustness means that you also include invalid values right next to the boundary.



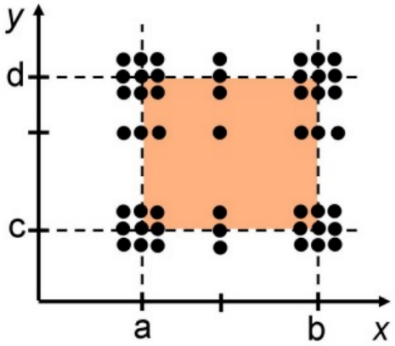
Adding worst-cases

- Worst-cases means that you include values right in the corner of the boundaries.



Adding robustness + worst-cases

- You can add them both to be extra secure.



Extras

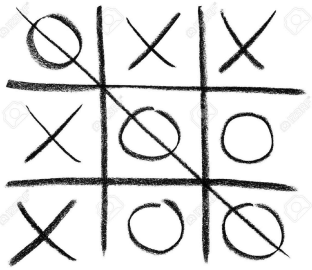
- There are other ones too, like:
    - **Special value testing**: using domain knowledge to add special values and finding corresponding boundaries for internal variables (white-box testing)
    - **Random testing**: randomly choosing inputs

Design patterns

Get-15 and Tic-tac-toe

- Get-15 is a game where two players pick numbers, and as soon as one person has a subset of numbers that adds up to 15, they win.

- Tic-tac-toe is a game where two players take turns to draw crosses and circles in a 3 x 3 grid, and the first one to make a line of their shape wins.



- These two games are actually identical, if you put Get-15 into a magic square (a square where all rows/columns add up to 15):

2	7	6
9	5	1
4	3	8

- Therefore, the same strategies for tic-tac-toe can be applied to Get-15 and vice versa.

Design patterns

- A design pattern is a reusable template for solving a recurring design problem based on design knowledge and is modifiable.

- A design pattern should have low coupling (modules can be replaced easily) and high cohesion (one module should do one job only), clear dependencies and explicit assumptions.
- Design patterns help by providing a shared vocabulary to designers and provide examples of modifiable designs.

- The four main elements of a design pattern are:
    - A unique *name*
    - A *problem description* that describes situations where the pattern can be used
    - A *solution* described as a set of collaborating classes and interfaces
    - A set of *consequences* describing trade-offs and alternatives relevant to the design goals in question

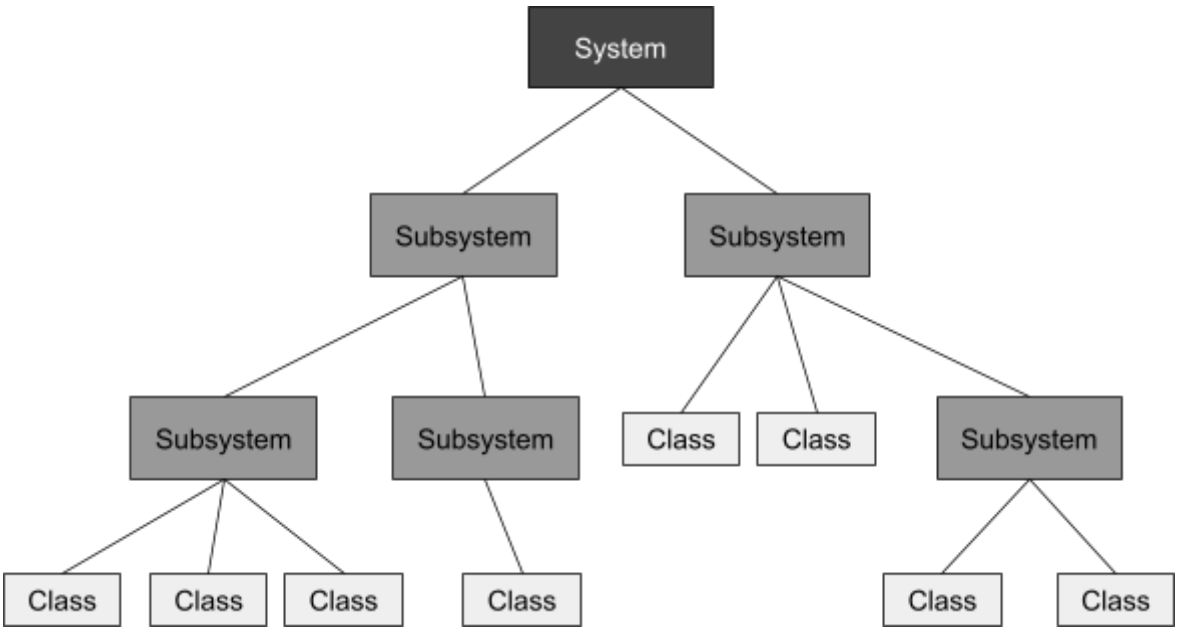
- The three types of design patterns are as follows:
    - **Structural** patterns (adapter, decorator etc.)
        - Reduce coupling between two or more classes
        - Introduce an abstract class to enable future extensions
        - Encapsulate complex structures
    - **Behavioural** patterns (observer, iterator, state etc.)
        - Allow a choice between algorithms and the assignment of responsibilities to objects (who does what)
        - Characterise complex control flows that are difficult to follow at runtime
    - **Creational** patterns (factory, singleton, prototype, builder etc.)
        - Allow to abstract from complex instantiation processes
        - Make the system independent from the way its objects are created, composed and represented.

Composite pattern

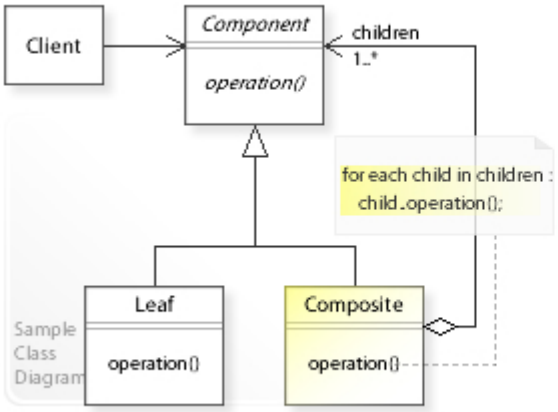
- **Software system**: A software system consists of subsystems which are either other subsystems or collection of classes
- **Software lifecycle**: A software lifecycle consists of a set of development activities which are either other activities or a collection of tasks

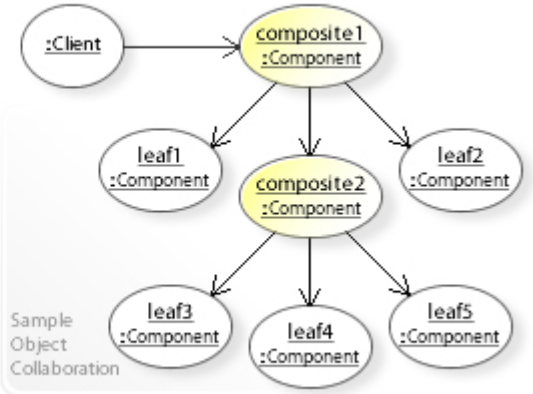
- These two sound similar. What's the same about them?
- They both follow the composite pattern, where a structure can be split up into two: a composite and a leaf.

- It's easier to visualise the composite pattern like a tree. Here is a tree representation of a software system:



- A more formal representation of the composite pattern is this:



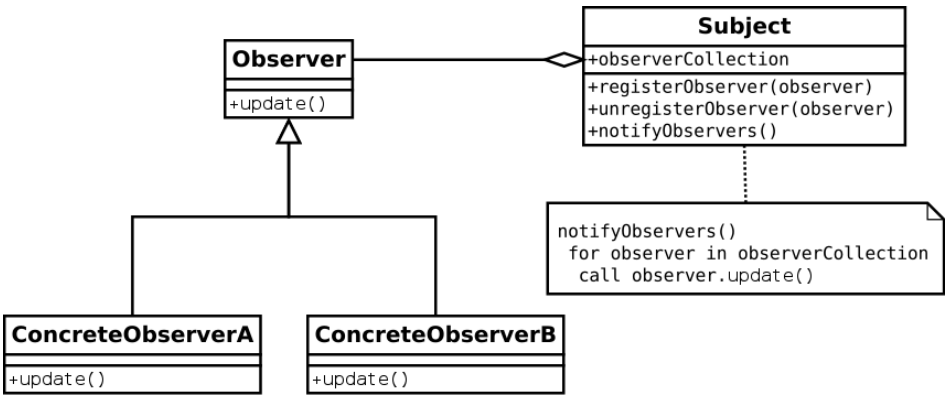


Taxonomies

- A taxonomy is simply a special kind of something, for example in a business, “Equipment”, “Facility”, “Fund” and “Organisation” are special kinds of “Resource”s and are therefore taxonomies.

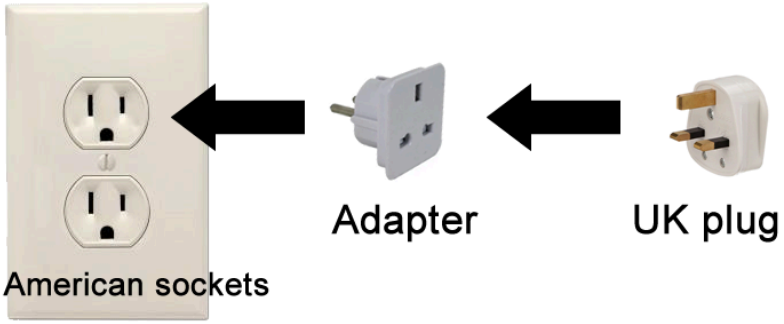
Observer

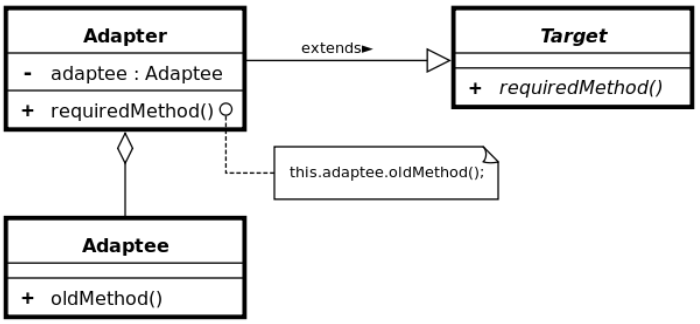
- An observer is a class that is registered with a subject.
- When the subject changes, the observer is notified and changes accordingly.
- The observer is part of the MVC architectural pattern, as the ‘view’.



Adapter

- An adapter pattern converts the interface of a class to another interface so it can work with other classes.
- Adapter patterns are mostly used to allow two incompatible classes to work together.
- It's very similar to its hardware counterpart, and it's easier to think of through using one as an example.
- Let's just say you travel to America, bringing some electronics from the UK. The plug in America is different from the one in the UK, so you can't plug it in. Therefore, you'd need an adapter to convert the UK plug for the American sockets.



- This same concept is used in code, and is called the adapter pattern.
- Naturally, this pattern only works for two pieces of code with the same base functionality (basically, they need to do the same thing, otherwise you can't adapt one to be like the other).

- The adapters can also be used as a "wrapper", which is a class where you put the original object in and the wrapper allows it to act like another interface.